

Ananya Penumudi

EMBEDDED & IOT ENGINEER

B.TECH CSE (IOT) · CLASS OF 2027

B.Tech CSE (IoT) student, graduating in **2027**, building **embedded systems, AI-powered devices and safety-critical software**. Industrial trainee at **BHEL Bengaluru** and author of a systems-engineering thesis on **KAVACH**, India's Automatic Train Protection system.



PORTFOLIO

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EDUCATION

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY

2023 – 2027 (expected)

B.Tech, Computer Science & Engineering (IoT) · JNTUH · CGPA 8.92 / 10 (till Semester VI)

SRI CHAITANYA JUNIOR COLLEGE

2021 – 2023

Intermediate, Telangana State Board · 97.4%

BHARATIYA VIDYA BHAVAN'S PUBLIC SCHOOL

2021

High School, CBSE · GPA 9.38 / 10

WORK EXPERIENCE

BHARAT HEAVY ELECTRICALS LIMITED (BHEL), BENGALURU

Jun – Jul 2026

Industrial Trainee → Internship certificate

- Trained at the Solar Business Division on manufacturing, production workflows, quality assurance and system integration in a Government of India PSU.
- Studied railway signalling and KAVACH through technical discussions and documentation.

ALFAGO

Jan 2026

Software Engineering Intern · Position Tracker → Internship certificate

- Built tax-lot based profit-and-loss calculations and backend transaction logic for a tracker covering transaction- and portfolio-level PnL.

KARTGEN INFOTECH LLP

Jan – Apr 2025

Operations Intern (remote) → Letter of recommendation

- Supported business development and operations: process coordination and stakeholder communication, under Shrivatsa Malviya.

FEATURED PROJECTS

ELEMENTIUM AI: VIRTUAL CHEMISTRY LAB

2026

Team project · React Three Fiber · FastAPI · MongoDB Atlas · Gemini → GitHub

- Built a 360° 3D lab with drag-and-drop apparatus, real-time titration and colour-change chemistry.
- Added a Gemini-powered tutor and a rule-based engine that catches unsafe or out-of-order steps; the lab notebook fills itself in.

BEYOND COLLISION PREVENTION: KAVACH THESIS

Jun 2026

Systems-engineering research · BHEL guidance → Thesis report → Simulator → Digital twin

- Analysed the architecture, operating modes, communication framework and functional safety of India's Automatic Train Protection system.
- Built an interactive simulator of KAVACH's modes of operation and a digital twin.

MORE PROJECTS

SMART AI WHEELCHAIR SYSTEM

Jan 2026

IoT + edge AI → Technical report

- Vision-based obstacle detection with automatic braking; GPS geofencing, caregiver alerts and sensor fusion across camera, IMU and proximity sensors.

AI-ENHANCED SECURE DATA WIPING PLATFORM

Nov 2025

AI + cybersecurity · team project → GitHub

- Binary-overwrite data sanitisation with cryptographic verification of every wipe and AI-based anomaly detection.

SYNCOPE: FALL DETECTION & ALERT SYSTEM

Aug 2025

Wearable IoT → Documentation

- A wearable that detects falls from motion-sensor data and alerts caregivers automatically.

FOCUSFLOW: ADHD-FRIENDLY PRODUCTIVITY PLATFORM

Apr 2025

AI + web → GitHub

- Mood tracking, journaling, facial-expression analysis and gamified focus tools.

AURALYN: EMOTIONAL-WELLNESS WEARABLE

2025 – now

IoT + AI + healthcare · in design

- Designing an ESP32-S3 + BLE wearable that pairs biometric sensing with mood analysis, with apps for the wearer, therapist and family.

EARLIER WEB PROJECTS

2023 – 2024

- Ideavoid**: startup-idea validation platform with peer feedback and market-viability assessment.
- Cheugy**: wellness app for stress management and self-care. → GitHub
- Syndicate**: gamified learning platform built around films, documentaries, quizzes and discussions.

TECHNICAL SKILLS

IOT & EMBEDDED	Arduino, ESP32, sensor integration (SPI, I2C, PWM), BLE, GPS, GSM, accelerometers
PROGRAMMING	C, C++, Python, Java, JavaScript, R, HTML, CSS, data structures & algorithms
AI & DATA	Computer vision, edge inference, mood detection, anomaly detection, Tableau
SYSTEMS ENGINEERING	Requirements analysis, functional safety, verification & validation, system architecture, traceability
WEB, CLOUD & TOOLS	React Three Fiber, FastAPI, MongoDB, MySQL, AWS, CI/CD, Figma, AutoCAD, StarUML
LANGUAGES	English, Hindi, Telugu

CERTIFICATIONS

- DevOps with Real-Time Practical Exposure**, Brainovision (AICTE approved), VNRVJIET · Jul – Nov 2025
- Tableau Certified Data Analyst Training**, Udemy (59 hours) · Jun 2024 · → Certificate
- RoboJAM Robotics Workshop**, Convergence 2K23, VNRVJIET · Dec 2023

AWARDS & ACTIVITIES

- 2nd Prize, IDEATHON 2023**, VNRVJIET · → Certificate
- Smart India Hackathon 2025**: solution design, prototyping and technical documentation
- 24-hour Hackathon, Convergence 2025, VNRVJIET · → Certificate
- 24-hour Hackathon DEMUX 2024, BVRIT · → Certificate
- National Conclave 2025, CBIT · → Certificate
- Member, Computer Society of India (VNRVJIET chapter) and Art of Living Cultural Club, 2023 – present